2011 VJC H1 Chemistry Prelim Exam 8872/2
Suggested Answers

Section A

Answer **all** questions in this section in the spaces provided.

1 (a) For decades, scientists were puzzled by why dinosaurs suddenly became extinct 65 million years ago. In studying core samples of rock dating back to that period, scientists found an unusual high level of iridium. This possibly came from an iridium-rich asteroid that struck the Earth's surface and the mystery to dinosaurs' extinction was solved.

Iridium exists as two naturally occurring isotopes, iridium-191 and iridium-193.

(i) Define the term *relative atomic mass*.

The relative atomic mass of an element is the ratio of the average mass of one atom of the isotope of the element to 1/12 the mass of an atom of ^{12}C isotope.

(ii) The relative atomic mass of iridium is 192.2. Calculate the natural abundance, in percentage, of each isotope.

Let the percentage abundance of ^{191}Ir be *x*, and
percentage abundance of ^{193}Ir is $(100 - x)$.
Ar of Ir = $(191x + 193(100-x))/100 = 192.2$
$191x + 193(100-x) = 192.2\ (100)$
$x = 40$

∴ percentage abundance of ^{191}Ir = 40 %
percentage abundance of ^{193}Ir = (100 – 40)% = 60%

(iii) Iridium is one of the rarest metals in the Earth's crust. It is thought to exist in 2 parts per billion (ppb). Calculate the mass of iridium (in kg), in 10 tonnes of Earth's crust.

[1 billion = 10^9, 1 tonne = 1000 kg]

Mass of iridium present in 10 ton Earth's crust

= 2 ppb of 10 ton Earth's crust

$$= \frac{2}{10^9} \times (10 \times 1000)\ \text{kg}$$

$= 2.0 \times 10^{-5}$ kg

[4]

(b) Aqua regia (a mixture of 75% nitric acid and 25% hydrochloric acid by volume) is highly corrosive. Only noble metals like iridium are inert to this solution. A 5 g sample of platinum-iridium alloy required 24.6 cm^3 of aqua regia for complete reaction. Platinum was completely oxidised to platinum(IV) ions by nitric acid and 0.5 g of metal was recovered.

(i) Find the percentage of each metal in the alloy.

Only Pt was oxidized during the reaction, Ir remains inert.
∴ 0.5 g of metal was recovered, the metal was Ir
percentage by mass of Ir in sample

$$= \frac{0.5}{5} \times 100$$

= 10%
Percentage by mass of Pt in sample
= (100 – 10)%
= 90%

(ii) The concentration of nitric acid used to make aqua regia is 5.0 mol dm^{-3}. Assuming that the reaction between the sample and aqua regia is complete, determine the amount of HNO_3 and Pt used in the reaction.

Volume of HNO_3 used
= 0.75 × 24.6
= 18.45 cm^3

No. of moles of HNO_3 reacted

$$= \frac{18.45}{1000} \times 5.0$$

= 0.09225 mol

No. of moles of Pt reacted

$$= \frac{4.5}{195}$$

= 0.02308 mol

(iii) Hence construct a balanced equation for the reaction between HNO_3 and Pt.
∴ mole ratio of HNO_3 : Pt = 4:1
∵ 4 mol of electron is transferred for every mol of Pt oxidized;

∴ 1 mol of electron is transferred for every mol of NO_3^- reduced.

$$NO_3^- + 2H^+ + e^- \longrightarrow NO_2 + H_2O \quad (1)$$
$$Pt \longrightarrow Pt^{4+} + 4e^- \quad (2)$$

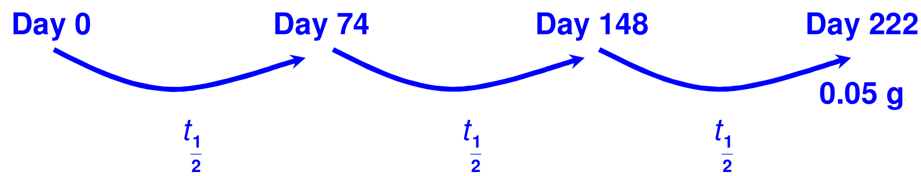
(1) × 4

$$4NO_3^- + 8H^+ + 4e^- \longrightarrow 4NO_2 + 4H_2O$$
$$\therefore 4NO_3^- + 8H^+ + Pt \longrightarrow 4NO_2 + 4H_2O + Pt^{4+}$$

[4]

(c) Man-made iridium-192 is an important radioactive isotope of iridium. It has a half-life of 74 days. The decay of a sample of iridium-192 was monitored.

(i) Calculate the original mass of the sample if the sample weighed 0.05 g after 222 days.



Let the original mass of the sample be *x* g.

$$x \times \left(\frac{1}{2}\right)^3 = 0.05$$

$$x = 0.40 \text{ g}$$

(ii) Suggest an application for iridium-192.

^{192}Ir can be used as a source of radiation for radio therapy in the treatment of cancer.

[2]

[Total: 10]

2 **Figure 1** shows the first eight ionisation energies of an element **X** while **Figure 2** shows the first ionisation energies of the eleven consecutive elements (including element **X**) from Period 2 and Period 3.

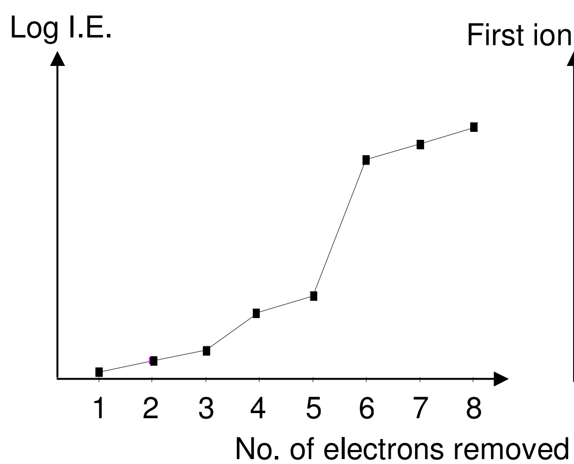


Figure 1

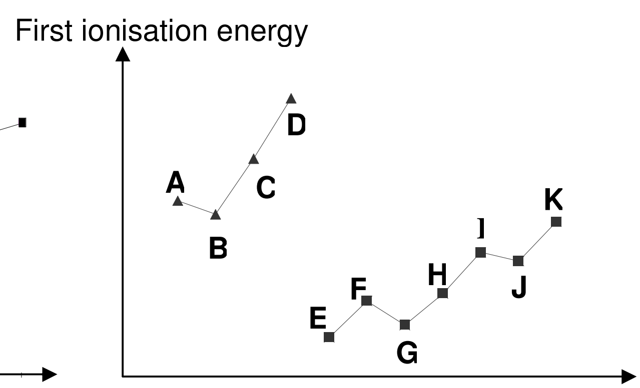


Figure 2

(a) (i) With reference to **Figure 1**, explain briefly the group to which element **X** belongs. . Hence, identify element **X.** [2]

Group V
A large difference in energy is required for removing 6^{th} electron, implying that the 5^{th} and 6^{th} electrons are removed from different shells, the 6^{th} electron comes from the inner shell and X has 5 valence electrons.

Since X has more than $7e^-$, hence it must be phosphorus, P.

(ii) Which letter in **Figure 2** corresponds to element **X**? Explain your answer briefly. [2]

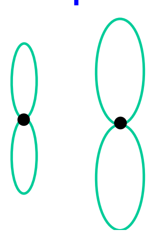
Element I.

P is in Period 3. A large drop in 1^{st} IE from D to E shows that elements E to K are in Period 3, with element E in Group I and element I in Group V.

[Or Two deviations in the first ionisation energy trend across the period, with the second deviation to be first IE of P > first IE of S.]

(iii) Write the electronic configuration of element **X** and draw two labelled diagrams of the two p orbitals of element **X** to illustrate their shapes and sizes. [2]

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$



2p 3p

(b) (i) Chlorides of elements **F** and **H** react with water separately to give solutions of pH values 6.5 and 2 respectively. Identify **F** and **H**. Hence account for the difference in pH values, with the help of equations [2]

F is Mg while H is Si.

$MgCl_2$ is ionic, dissolves in water to form Mg^{2+} and Cl^-. Due to the high charge density of Mg^{2+}, it hydrolyses to give slightly acidic solution.

$MgCl_2 + H_2O \rightleftharpoons MgCl(OH) + HCl$

$SiCl_4$ is a simple covalent molecule which hydrolyses fully in water to give a strongly acidic solution.
$SiCl_4(l) + 2H_2O(l) \rightarrow SiO_2(s) + 4HCl(aq)$

(ii) The acid-base nature of oxides of **E** and **I** are different. Illustrate this property by writing balanced equations to show their reactions, if any, with aqueous sodium hydroxide and aqueous hydrochloric acid.

[2]

E is Na while I is P.
Na_2O is <u>basic</u> in nature while the P_4O_{10} is <u>acidic</u> in nature.
$Na_2O + 2HCl \rightarrow 2NaCl + H_2O$
$P_4O_{10} + 12NaOH \rightarrow 4Na_3PO_4 + 6H_2O$
[or $P_4O_6 + 8NaOH \rightarrow 4\ Na_3PO_3 + 2H_2O$]

[Total: 10]

3 (a) A United Nations toxicologist studying the properties of mustard gas, $S(CH_2CH_2Cl)_2$, a blistering agent used in warfare, mixes 50 cm^3 of 1.34 mol dm^{-3} SCl_2 with 50 cm^3 of 1.96 mol dm^{-3} C_2H_4, and allows it to react at 20°C. At equilibrium, the concentration of $S(CH_2CH_2Cl)_2$ is found to be 0.350 mol dm^{-3}.

$$SCl_2\,(alc) \quad + \quad 2C_2H_4\,(alc) \rightleftharpoons S(CH_2CH_2Cl)_2\,(alc)$$

Give the expression for K_c and calculate its value for the above reaction at 20°C.

[3]

	SCl_2 (g)	**$2C_2H_4$ (g)**	**$S(CH_2CH_2Cl)_2$ (g)**
[Initial] mol dm^{-3}	**½ x 1.34 = 0.670**	**½ x 1.96 = 0.980**	**0**
[Change] mol dm^{-3}	**-0.350**	**-0.700**	**+0.350**
[Equilibrium] mol dm^{-3}	**0.320**	**0.280**	**0.350**

$$K_c = \frac{[S(CH_2CH_2Cl)_2]}{[SCl_2]\,[C_2H_4]^2}$$

$$= 0.350/(0.320)(0.280)^2$$

$$= \underline{14.0\ mol^{-2}\ dm^6}$$

(b) Explain the terms *order of reaction* and *half-life*. [2]

Consider the <u>experimentally determined</u> rate equation: Rate = $k[A]^x[B]^y$
<u>x</u> is the order of reaction <u>with respect to reactant A</u> while
<u>y</u> is the order of reaction <u>with respect to reactant B</u> and
<u>(x+y)</u> is the <u>overall order</u> of the reaction.

Half-time is the time taken for <u>half</u> the <u>amount of reactant present to react</u>.

(c) Hydrogen peroxide reacts with acidified iodide ions to give iodine. The following results were obtained in a study of the kinetics of this reaction.

Experiment	Initial concentrations of reactants/ mol dm^{-3}			Initial rate of formation of iodine/ mol dm^{-3} s^{-1}
	$[I^-]$	$[H_2O_2]$	$[H^+]$	
1	0.010	0.020	0.10	6.0×10^{-6}
2	0.010	0.030	0.20	9.0×10^{-6}
3	0.020	0.030	0.10	1.8×10^{-5}

(i) Write a balanced equation for the reaction. [1]

$H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$

(ii) The reaction is zero order with respect to hydrogen ions. Determine the order of reaction with respect to iodide and hydrogen peroxide. [2]

Order of reaction with respect to iodide : <u>1</u>

Working:
Consider experiments 2 and 3, keeping $[H_2O_2]$ constant, the change in $[I^-]$ is the same as the change in initial rate.
Hence, order of reaction wrt to I^- = 1

Order of reaction with respect to hydrogen peroxide : <u>1</u>

Working:
Consider experiments 1 and 2, keeping $[I^-]$ constant, the change in $[H_2O_2]$ is the same as the change in initial rate.
Hence, order of reaction wrt to H_2O_2 = 1

(iii) Write a rate equation for this reaction. [1]

Rate = $k [I^-] [H_2O_2]$

(iv) Hence, calculate a value for the rate constant, giving its units. [1]

[1 for correct ans and units; ½ mark for no or wrong units;
0 mark for wrong value with right or wrong units]

Using data from experiment 1,
$k = 6 \times 10^{-6} / (0.01 \times 0.02)$ = <u>3.00×10^{-2} mol^{-1} dm^{3} s^{-1}</u>

[Total: 10]

4(a) Succinic acid (butanedioic acid) is widely used in the food and beverage industry. Industrially it is produced from the fermentation of carbohydrates. A student proposes the following synthetic route for the synthesis of succinic acid from butane:

$$CH_3CH_2CH_2CH_3 \xrightarrow[\textbf{Step 1}]{Cl_2\text{, uv light}} ClCH_2CH_2CH_2CH_2Cl$$

$$ClCH_2CH_2CH_2CH_2Cl \xrightarrow[\textbf{Step 2}]{\text{Aq. NaOH, reflux}} HOCH_2CH_2CH_2CH_2OH$$

$$HOCH_2CH_2CH_2CH_2OH \xrightarrow{\textbf{Step 3}} \underset{\text{Succinic acid}}{HOOCCH_2CH_2COOH}$$

(i) Suggest the reagents and conditions for step 3.

H_2SO_4/ $K_2Cr_2O_7$ or $KMnO_4$; reflux [1]

(ii) Name the type of reaction involved in the first step. Hence comment on the yield of acid obtained.

Substitution.
Yield would be <u>low</u> as <u>substitution is not specific</u>.

[2]

(iii) Suggest an alternative route for synthesizing succinic acid starting from ethene. State clearly all reagents and conditions required. [5]

$$\mathbf{CH_2{=}CH_2 \xrightarrow[]{Br_2 \text{ in } CCl_4,\ rtp} CH_2BrCH_2Br \xrightarrow[\text{reflux}]{\text{Alc KCN/ NaCN}} CH_2(CN)CH_2(CN)}$$

$$\mathbf{CH_2(CN)CH_2(CN)} \xrightarrow[\textbf{heat}]{\textbf{Dil. HC}\boldsymbol{l}\textbf{ / }\mathbf{H_2SO_4}} HOOCCH_2CH_2COOH$$

(iv) Calculate the volume of gas evolved at rtp when an excess of sodium metal is added to 60 g of the succinic acid. [2]

$$\mathbf{HOOCCH_2CH_2COOH + 2Na \longrightarrow NaOOCCH_2CH_2COONa + H_2}$$

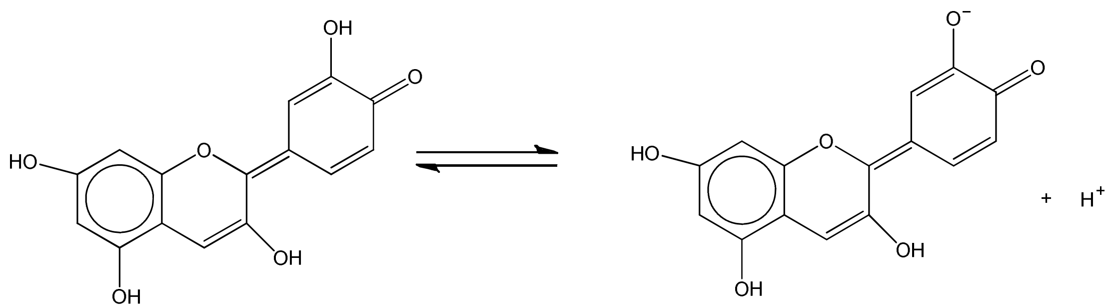
Number of moles of acid used = 60/118 = 0.508 mol

Number of moles of gas released = 0.508 mol

Volume of gas at rtp = 0.508 x 24.0 = <u>12.2 dm^3</u>

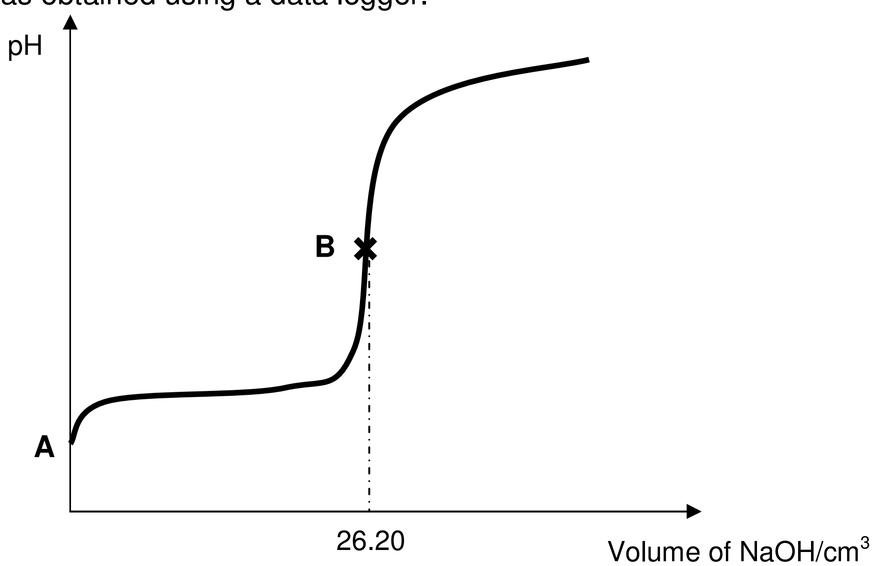
Section B

5 Cyanidin is a natural pH indicator that can be found in many red berries, including grapes, blackberry and cherry. Aqueous solutions of cyanidin changes colour with pH, appearing red at pH < 3, violet at pH 7-8, and blue at pH > 11. In addition, cyanidins are antioxidants that can scavenge free radicals and counteract ageing caused by oxidative damage in our body tissues and cells.



Cyanidin (CyH)

Cyanidin (CyH) is a *weak monoprotic acid*. In one experiment, 25.0 cm^3 of cyanidin was titrated against 0.10 mol dm^{-3} of dilute NaOH at 25℃. The following titration curve was obtained using a data logger.



(a)(i) Explain what is meant by a *weak monoprotic acid*. [1]

A weak acid is an acid that <u>dissociates partially.</u> A monoprotic acid when dissociated completedly donates only <u>one proton</u> per acid molecule.

$$\text{i.e. } CyH \rightleftharpoons Cy^- + H^+$$

(ii) Calculate the molar concentration of the cyanidin solution. [1]

Since $n_{CyH} = n_{NaOH}$,

Conc. of cyanidin = $\frac{0.10 \times 26.20}{25.0}$ **=** **$\underline{0.105\ mol\ dm^{-3}}$**

(iii) Given that a solution of cyanidin is only 0.8% dissociated into ions, calculate the pH at point **A**. [1]

$[H^+] = [Cy^-] = 0.008 \times 0.105 = 8.38 \times 10^{-4}\ mol\ dm^{-3}$
$pH = -\log_{10}[H^+] = \underline{3.08}$

(iv) Write an expression for the acid dissociation constant, K_a, for cyanidin. Hence, calculate the value of K_a for cyanidin, stating clearly its units. [2]

$$K_a = \frac{[Cy^-][H^+]}{[CyH]}$$

$$K_a = \frac{(8.38 \times 10^{-4})^2}{0.105} = \underline{6.69 \times 10^{-6}\ mol\ dm^{-3}}$$

(v) Explain, with the aid of an appropriate equation, why the pH at end-point **B** is greater than 7. [2]

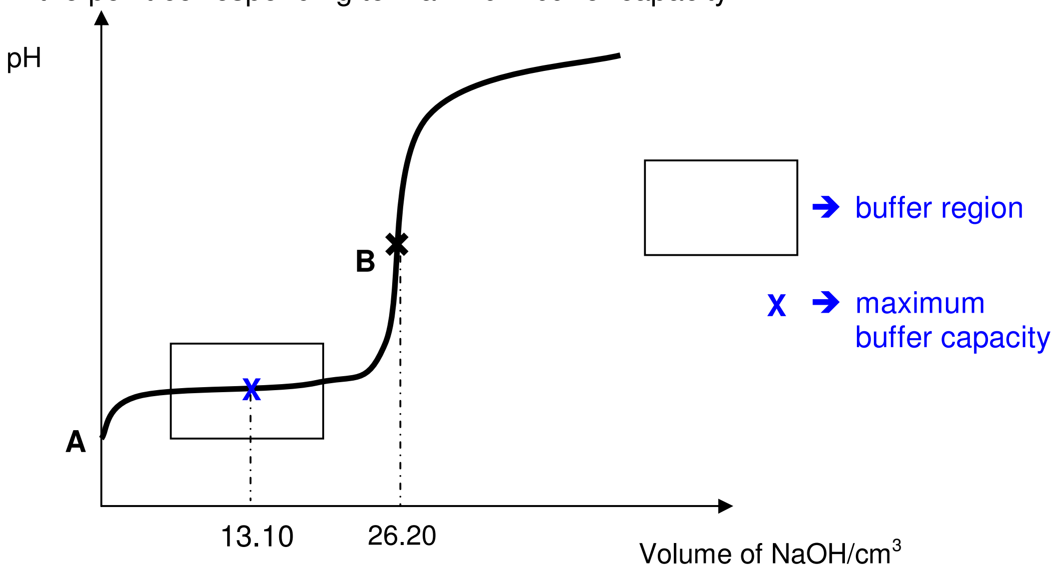
At end-point B, only the conjugate base of the weak acid, Cy^- is present.
$Cy^- + H_2O \rightleftharpoons CyH + OH^-$
Cy^- __hydrolyses__ in water to produce __excess OH^- ions__. Hence, __pH > 7__.

(b) An aqueous solution of cyanidin CyH and Cy^- can act as a buffer.

(i) Copy the titration curve onto your answer script and label

I –the buffer region

II–the point corresponding to maximum buffer capacity.



[2]

(ii) Explain, with the help of equations, how an aqueous mixture of CyH^+ and Cy^- can control pH when relatively small amounts of acid or base is added to the solution. [3]

When a small amount of acid is added,

$$Cy^-(aq) + H^+(aq) \rightarrow CyH(aq)$$

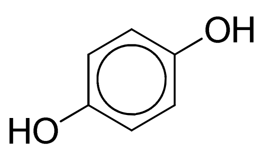
Large reservoir of Cy^- ions from the salt in the buffer remove the additional H^+ ions and the pH of the solution remains almost constant.

When a small amount of base is added,

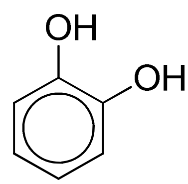
$$CyH(aq) + OH^-(aq) \rightarrow Cy^-(aq) + H_2O(l)$$

Large reservoir of CyH molecules in the buffer remove the additional OH^- ions and the pH of the solution remains almost constant.

(c) Cyanidin can be synthesized via a number of steps from either of the two molecules shown below:



Molecule P



Molecule Q

(i) Which molecule **P** or **Q** will have a higher boiling point? Explain your answer. [2]

Molecule P will have a higher boiling point as more energy will be required to overcome the stronger inter-molecular hydrogen bonding in P. On the other hand, due to the proximity of the –OH groups in molecule Q, intra-molecular hydrogen bonding predominates. As such, less energy is required to break the weaker dipole-dipole interactions in Q.

(ii) **P** dissolves readily in water at room temperature but two immiscible layers were observed when P was added to trichloromethane.
Account for the above observation. [3]

In P, the H is directly bonded to highly electronegative O, hence it is able to form hydrogen bonding with water, which is sufficient to overcome the solute-solute interaction, i.e. hydrogen bonding between P molecules and the solvent-solvent interaction, i.e. hydrogen bonding between water molecules, hence P is soluble in water.
However, P forms only weaker dipole-dipole interaction with trichloromethane, which is insufficient to overcome the stronger hydrogen bonding between P molecules and the dipole-dipole interaction between polar trichloromethane molecules.

(d) Predict and explain the relative acidity of propanoic acid and 2-chloropropanoic acid. [3]

$CH_3CHClCOOH \rightleftharpoons CH_3CHClCOO^- + H^+$ ....(1)

$CH_3CH_2COOH \rightleftharpoons CH_3CH_2COO^- + H^+$

$CH_3CHClCOO^-$ is <u>more stable</u> than $CH_3CH_2COO^-$ as the negative charge on the carboxylate ion is <u>further dispersed by the electron-withdrawing chlorine atom</u>. Hence equilibrium (1) lies more to the <u>right</u> hence <u>2-chloroprppanoic acid is a stronger acid</u> than propanoic acid.

[Total: 20]

6 Nitric acid, HNO_3, known to early chemists as aqua fortis (meaning "strong water"), is a major commercial acid. It is available as the concentrated acid which is a 70% aqueous solution. Although colourless when pure, the acid turns pale brown when exposed to light as a result of decomposition.

$$4HNO_3(aq) \longrightarrow 4NO_2(g) + 2H_2O(l) + O_2(g) \text{ ....}\mathbf{(1)}$$

Pure nitric acid can be prepared by distillation from a mixture of potassium nitrate and concentrated sulfuric acid as shown

$$KNO_3 + H_2SO_4 \longrightarrow KHSO_4 + HNO_3 \text{ ....}\ \mathbf{(2)}$$

This method was at one time used in the commercial preparation of nitric acid, but it has now been replaced by the Oswald process, discovered in 1902 by the German chemist Wilhelm Ostwald, who received the Nobel Prize in chemistry in 1909 for his work on catalysts. In the Ostwald process, ammonia gas is burned in oxygen over a platinum catalyst. The product is nitric oxide, NO, which reacts with oxygen to give nitrogen dioxide. When dissolved in water, NO_2 produces nitric acid. The equations are

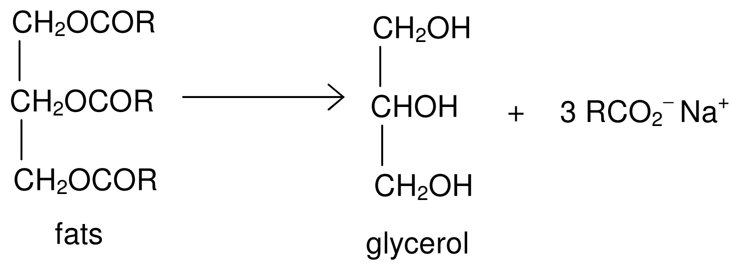
$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g); \quad \Delta H_1 = -906 \text{ kJ mol}^{-1}$$

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g); \quad \Delta H_2 = -114 \text{ kJ mol}^{-1}$$

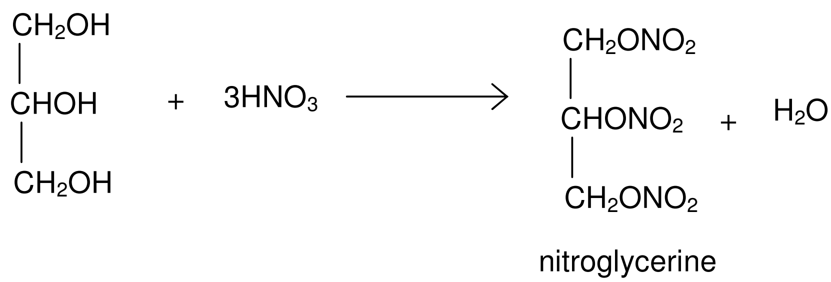
$$3NO_2(g) + H_2O(l) \longrightarrow 2HNO_3(aq) + NO(g); \quad \Delta H_3 = -137 \text{ kJ mol}^{-1}$$

The largest single use of nitric acid is in the production of ammonium nitrate, NH_4NO_3, which is largely used as an agricultural fertiliser. It is also used to manufacture chemicals for nylon and to make explosives like nitroglycerine, $C_3H_5(NO_3)_3$. In making the explosive nitroglycerine, glycerol, $C_3H_5(OH)_3$, which is obtained from fats is used as the starting material. The manufacturing process is shown in two stages as shown below:

Stage 1



Stage 2



(a) Explain why nitric acid in a bottle often turns pale brown after some time.

It turns pale brown when exposed to light due to production of NO_2 gas which appears pale brown at a lower concentration.

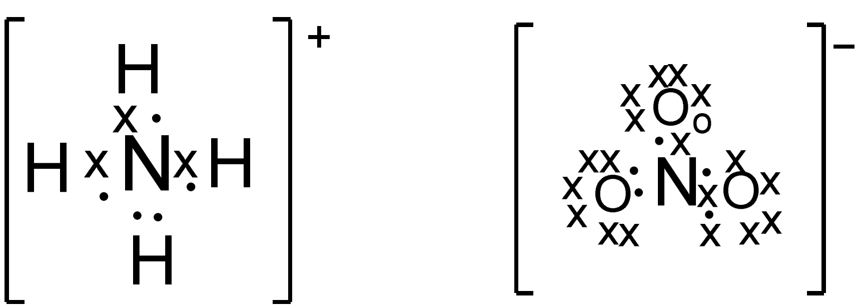
[1]

(b) State the type of reaction involved in the making of pure nitric acid as exemplified in reaction **(2)**. Explain your answer.

Acid-base reaction.
Sulfuric acid acts as acid (proton donor) whereas nitrate acts as base (proton acceptor).

[2]

(c) Draw the dot and cross diagrams for the ions in NH_4NO_3. Hence, predict the shape of each ion present.



tetrahedral **trigonal planar**

[4]

(d) **(i)** Calculate the average bond energy of the bond found in the NO molecule formed in the following reaction, using relevant data from the *Data Booklet*.

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g); \qquad \Delta H_1 = -906 \text{ kJ mol}^{-1}$$

Total energy absorbed in breaking bonds in reactants
= 12(390) + 5(496) = 7160 kJ

Total energy released in forming bonds in products
= 4[BE(NO)] + 12(460) = {4[BE(NO)] + 5520} kJ

$\Delta H = -906 = 7160 - \{4[BE(NO)] + 5520\}$
BE(NO) = 636.5 kJ mol^{-1}

[3]

(ii) Suggest a reason for the discrepancy between your answer in part **(d)(i)** and the experimentally determined value of of 629 kJ mol^{-1}.

The values given in the Data Booklet are <u>average</u> bond energy values.

[1]

(e) Using appropriate enthalpy change values given for the Ostwald Process and the following energy cycle, calculate the enthalpy change for reaction **(1)**.

$$\begin{tikzcd}
2HNO_3(aq) + NO_2(g) \arrow[rr, "\triangle H_1 / 2"] & & 3NO_2(g) + H_2O(l) + \tfrac{1}{2}O_2(g) \\
& 2HNO_3(aq) + NO(g) + \tfrac{1}{2}O_2(g) \arrow[ul, "-114/2 \text{ kJ mol}^{-1}"] \arrow[ur, "+137 \text{ kJ mol}^{-1}"'] &
\end{tikzcd}$$

$\triangle H_1 / 2 = -(-114/2) + (+137)$
$\triangle H_1 = +388$ kJ mol^{-1}

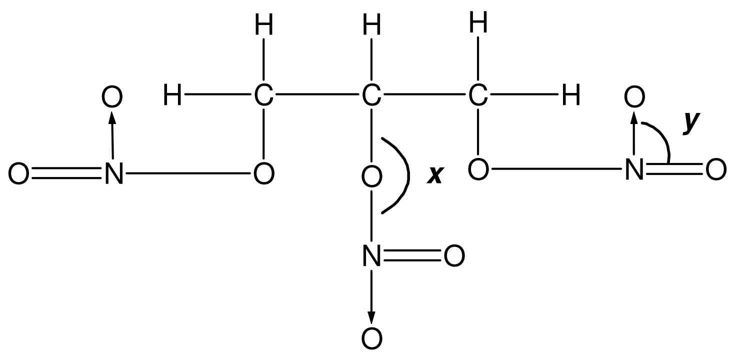
[2]

(f) State the reagents and conditions used for the formation of glycerol from fats.

Aqueous NaOH, reflux
Or Aqueous HCl, reflux

[1]

(g) The molecule of glycerine has the following structure:



State the C–O–N bond angle (***x***) and the O–N–O bond angle (***y***) as indicated in the diagram above.

y = Bond angle of ONO = 120^{o} (3 bond pairs / 0 lone pair)
x = Bond angle of CON = 105^{o} (2 bond pairs / 2 lone pairs)

[2]

(h) Nitroglycerine is hazardous to handle in the pure state. It is very sensitive to shock, as shown by the following reaction:

$$4C_3H_5(ONO_2)_3(l) \longrightarrow 6N_2(g) + 10H_2O(g) + 12CO_2(g) + O_2(g);$$
$$|\Delta H_4| = 5.720 \times 10^3 \text{ kJ mol}^{-1}$$

Alfred Nobel (1833 -1896) discovered that when nitroglycerine was mixed with clay and similar materials, it was much less shock-sensitive. The mixture is called dynamite. One stick of dynamite weighs 500 g and contains 80% of nitroglycerine by mass.

(i) Predict the sign of ΔH_4 and explain your answer.

The sign is <u>negative</u>
due to formation of <u>strong triple bonds</u> in the nitrogen molecules making ΔH_4 <u>highly exothermic</u>.

(ii) Dynamite is still widely used in many countries for clearing of obstacles and for mining resources. However, its potential danger of explosion cannot be underestimated.
Calculate the average change in temperature of the surrounding 20 m^3 of air around **one** stick of exploding dynamite.

[Volumetric heat capacity of air, C = 1.297 kJ m^{-3} K^{-1}]

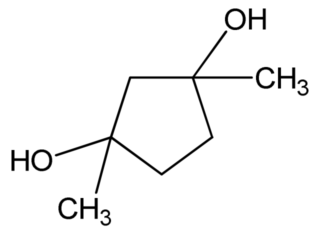
$\Delta T = \{[(1\times 500 \times 0.8)/227] \times (5.720 \times 10^3 /4 \} / (1.297 \times 20)$
= <u>97 K</u>

[4]

[Total: 20]

7 (a) Suggest methods by which the following pairs of compounds could be distinguished from each other by chemical tests. State clearly how each compound would behave. For each case, **no more than two steps** should be used.

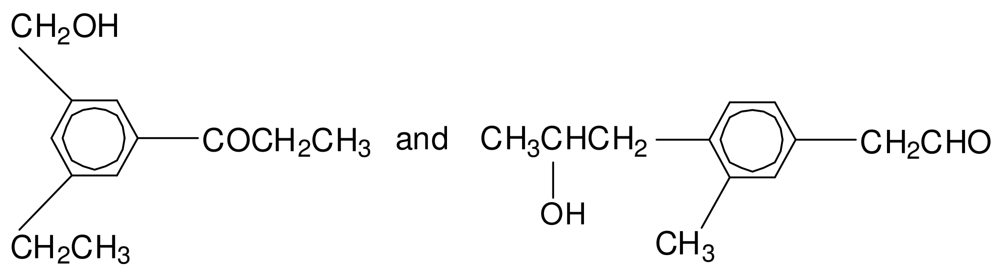
(i)



and $CH_3(CH_2)_5CO_2H$

Add aqueous sodium carbonate at room temperature.
The 2nd compound gives CO_2 effervescence while the 1st compound does not.

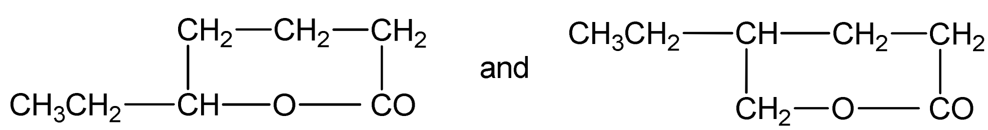
(ii)



Add Fehling's solution and warm. [OR Tollens' reagent and warm]
The 2nd compound gives a brick red ppt [OR silver mirror] while the 1st compound gives no ppt.

(alternative iodoform test)

(iii)



[7]

Add acidified $Cr_2O_7^{2-}$(aq) [OR MnO_4^-] and heat. [Ester will be hydrolysed and oxidation of alcohol takes place.]
Add 2,4-dinitrophenylhydrazine at room temperature to the resulting product mixture.
1st compound will give an orange ppt whereas 2nd compound gives no ppt.

(b) **A** ($C_4H_8Cl_2$) reacts with sodium hydroxide under different conditions. In aqueous medium, **A** forms **B** ($C_4H_{10}O_2$) while in ethanolic medium, **A** forms the major product **C** (C_4H_7Cl). 1 mol of **C** reacts with 1 mol of aqueous bromine.

B reacts with warm alkaline aqueous iodine to give a pale yellow precipitate as well as **D** ($C_3H_5O_3Na$). Both **C** and **D** undergo vigorous oxidation in acidic medium to produce **E** and **F** respectively.

0.0500 mol of **F** was dissolved in water and the solution was made up to 250 cm^3. A 25.0 cm^3 portion of this solution required 13.30 cm^3 of 0.75 mol dm^{-3} NaOH for complete reaction.

When **E** is refluxed with alcoholic KCN, the product formed has the molecular formula $C_3H_3NO_2$.

(i) Deduce the structures of all the lettered compounds (**A** to **F**) and explain the reactions involved.
(ii) Write an equation to represent the conversion of **A** to **C**.

[13]

Evidence	**Deduction**
A + aq NaOH → B	**A undergoes substitution with aq. NaOH.** **B is an alcohol.**
A + alc NaOH → C **C + 1 mol Br_2**	**A undergoes elimination with alc. NaOH.** **C undergoes addition and has 1 C=C.**
B + NaOH / I_2 **→ yellow ppt + D**	**B undergoes oxidative cleavage.** **B has –CH(OH)CH_3 group.** **Yellow ppt is CHI_3.** **D is a carboxylate salt.**
D oxidized to F **Titration of F with NaOH (aq)**	**–CO_2H in F is neutralized by NaOH.** **No. of moles of F = (25 / 250) × 0.0500 = 0.00500** **No. of moles of NaOH** **= (13.30 / 1000) × 0.75 = 0.00998** **n_{OH^-} / n_F = 0.00998 / 0.005 = 2** **F has 2 –CO_2H groups,** **i.e. D has 1 primary alcohol group that is oxidized to give an additional –CO_2H.**
C oxidized to E **E + KCN → $C_3H_3NO_2$**	**Since there is 1N in C_3H_3NO, 1 –Cl atom in E has been substituted by KCN to form nitrile.** **Hence, E has 2 C atoms.**

Structures

A

$ClCH_2—CH_2—CHCl—CH_3$

B

$HOCH_2—CH_2—CH(OH)—CH_3$

C

$ClCH_2—CH{=}CH—CH_3$

D

$HOCH_2—CH_2—CO_2^-Na^+$

E

$ClCH_2—CO_2H$

F

$HO_2C—CH_2—CO_2H$

(ii) Eqn

$$ClCH_2—CH_2—CHCl—CH_3 + NaOH \longrightarrow ClCH_2—CH{=}CH—CH_3 + NaCl + H_2O$$